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From visual features to key concepts: A Dynamic and Static Concept-driven approach for video captioning

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ABSTRACT

In video captioning, accurately identifying and summarizing key concepts while ignoring irrelevant details remains a significant challenge. Mainstream approaches often suffer from the inclusion of semantically irrelevant features, leading to inaccuracies and hallucinations in the generated captions. This study aims to develop a novel framework, **D**ynamic and **S**tatic **Co**ncept-driven video captioning model(DiSCo), to enhance the accuracy and coherence of video captions by effectively leveraging pre-trained models and addressing the issue of semantic irrelevance. DiSCo builds upon the conventional encoder–decoder architecture by incorporating a Semantic Feature Extractor (SFE) and a Static-Dynamic Concept Detector (S-DCD). The SFE filters out semantically irrelevant features extracted by the visual model, while the S-DCD identifies critical concepts to guide the large language model (LLM) in generating captions. Both the visual model and the LLM are pre-trained and their parameters are frozen; only the SFE and S-DCD are trained to optimize the feature extraction and concept detection processes. Comprehensive experiments conducted on the MSVD and MSR-VTT datasets show that DiSCo significantly outperforms existing methods, achieving notable improvements in the quality and relevance of the generated captions. The proposed DiSCo framework demonstrates a robust solution for enhancing the accuracy and coherence of video captions by effectively integrating semantic feature extraction and concept-driven guidance.

1. Introduction

Video captioning is the task of automatically generating brief yet informative textual descriptions for videos. This interdisciplinary subfield combines computer vision and natural language processing to process spatiotemporal information, align it with text, and generate captions. Recent advancements [1–5] in deep learning have propelled video captioning research towards a predominance of encoder–decoder frameworks, leveraging sophisticated models from computer vision and natural language processing (NLP). However, the direct application of these pre-trained models to video captioning encounters significant challenges: (1)Task Mismatch. Visual encoders [6,7], originally trained for video retrieval or action recognition, struggle to provide the nuanced detail required for generative tasks like video captioning. Their proficiency in recognizing broad actions and retrieving visuals does not translate seamlessly to the detailed scene understanding needed for comprehensive caption generation. (2)Modal Discrepancy. Integrating CV and NLP for video captioning faces the inherent challenge of bridging two vastly different domains. Videos encapsulate rich, multifaceted information, including repetitive or less critical elements that can introduce noise, potentially resulting in misunderstandings by the language model and impacting its interpretative accuracy. It is imperative to refine the encoder's output by filtering out irrelevant concepts and features to enhance the relevance and accuracy of generated captions.

In this paper, we present a Dynamic and Static Concept-driven video captioning model(DiSCo), which is an LLM-based end-to-end model for video captioning that effectively utilizes pre-trained models. Inspired by human behavior—wherein viewers focus on core, persistent concepts, and neglect ancillary details, DiSCo innovates by utilizing concepts to help training instead of traditional fine-tuning. This approach allows for the identification and emphasis of critical concepts during training, enabling the model to filter and refine semantically relevant features while discarding extraneous visual information. Furthermore, concepts as hints ensures that DiSCo accurately captures the essence of the content and effectively avoids generating off-topic descriptions. With concepts as the driving force, DiSCo efficiently bridges the pre-trained visual model and the LLM without necessitating fine-tuning of these pre-trained models, thus excelling at video captioning tasks while significantly reducing the training cost.

In our approach, the encoder adopts a two-stream architecture, including an RGB branch and optical flow branch, to extract static

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features and dynamic features respectively, to represent video features more comprehensively. To bridge the video-text disparity and utilize crucial concepts, DiSCo introduces a Semantic Feature Extraction (SFE) module and a Static-Dynamic Concept Detection (S-DCD) module. The SFE module refines raw visual features from encoded video frames. Meanwhile, the S-DCD module utilizes these refined features to anticipate relevant concepts, serving as informative cues for the decoder. Distinct from conventional approaches, DiSCo bypasses the creation of dataset-specific vocabularies for classification, opting instead to estimate word embeddings directly. Our decoder, powered by Large Language Models (LLMs), integrates both semantic and concept features through a carefully crafted prompt. Here, semantic features take precedence as the primary input, complemented by concept features acting as guidance. Leveraging the vast knowledge base and superior reasoning capabilities of LLMs, concept prompts play a pivotal role in steering caption generation. Notably, the synergistic effect of S-DCD in aiding SFE to distill crucial features and align them with textual representations obviates the need for encoder and decoder finetuning. This design dramatically curtails overall training duration, showcasing DiSCo's efficiency in video captioning while maintaining high accuracy. In summary, our main contributions are as follows:

(1) We present a Dynamic and Static Concept-driven video captioning model(DiSCo), a concept-driven, LLM-based end-to-end framework for video captioning. It pioneers cross-modal synergy via concept prompting, adeptly leveraging pre-trained models' inherent knowledge for video captioning without necessitating fine-tuning.
(2) Semantic feature extractor(SFE) is proposed to distill critical features from the visual representations of the pre-trained encoders, thus enabling adaptation to video captioning tasks and enhancing the semantic relevance of these features.
(3) Static-dynamic concept detector(S-DCD) is proposed to drive SFE to focus on extracting pertinent semantic features and predict key concepts to guide LLM to concentrate on core concepts to reduce hallucination.

2. Related works

2.1. Video captioning

Originating from seminal work [8], the encoder–decoder architecture has solidified its dominance in video captioning. This paradigm entails encoding individual video frames to extract features, which are then processed by a decoder. Building upon this foundational structure, research efforts have predominantly focused on two core dimensions. First, enhancing video representation: [9] extracted features across multiple levels. Second, bridging the gap between visual and textual modalities, where [10] leveraged high-level semantics for enhanced video-text interaction.

2.2. Video representation

Video is unique compared to pictures in that it contains not only static content (e.g., scenes, objects, and people), but also dynamic content (e.g., events), so spatio-temporal modeling is necessary. Researchers have focused on exploring different 2D/3D visual representations [6,7,11]. Since video captioning requires a deep understanding of open word semantics, existing video captioning [4,10,12,13] used features extracted based on models pre-trained on large-scale action recognition datasets. However, for many video tasks (such as action recognition) and datasets, given a single frame of video, existing models trained on image datasets can achieve high performance, even on par with models using multiple frames. The strong performance of single video frames indicates that the video representation is biased towards static appearance information.

2.3. Concept-augmented visual captioning

Numerous research efforts have been dedicated to identifying and leveraging concepts in images or videos to enhance the performance of caption generation. The pioneering work [14] lays the groundwork for concept-augmented visual captioning by selecting a predetermined set of concepts from image captions within the training data, subsequently designing a tailored concept detection network that employs semantic attention to steer caption creation. Subsequent studies [15,16] also frequently employ concept detection strategies but differ significantly in their application methods. In contrast to static images, video captioning presents a multi-modal challenge, thus necessitating distinct input configurations for concept detection. To tackle this complexity, [17] incorporated audio modalities to enrich the feature representation, yet their approach lacked a training objective for extracting key concepts. On the other hand, [18] augmented optical flow information to improve detection accuracy, but their method still required extracting a fixed set of concepts from the dataset, thus lacking generalizability.

3. Methods

Fig. 1 shows the framework of our model. Based on the encoder–decoder architecture, our model aims to comprehensively characterize videos and narrows the gap between video and text driven by concepts. Next, we propose a baseline model for video captioning and detail the semantic feature extractor and the static-dynamic concept detector module.

3.1. Video captioning baseline

Without any auxiliary modules, we introduce a naive encoder–decoder baseline here. In the encoding stage, given a video frame(or optical flow) sequence $V = \{v_1, v_2, \dots, v_N\}$ of length N and a feature dimension d_E for the encoder and d_D for the decoder. V is transformed into a feature vector $F_v \in \mathbb{R}^{N\times d_E}$ by an encoder and then transformed into an input embedding $E_v \in \mathbb{R}^{N\times d_D}$ by a linear layer. The formula can be defined as follows.

$$F_v = Encoder(V), \tag{1}$$

$$E_v = FC(F_v), \tag{2}$$

where F_v is the output of the encoder, E_v is the input embedding of the decoder, and $FC(\cdot)$ is the fully connected layer. If optical flow features are required, utilize two distinct encoders to acquire RGB feature F_v^{RGB} and optical flow feature F_v^{OPT}, subsequently fused to generate video feature $F_v \in \mathbb{R}^{(N_{RGB}+N_{OPT})\times d_E}$.

In the decoding stage, we design a prompt to combine with video features due to the adoption of LLM as decoder. First, given the prompt text P, it is transformed into a feature vector E_p after embedding.

$$E_p = Embedding(P), \tag{3}$$

where $Embedding(\cdot)$ is text embedding of the LLM.

Then, the embedding of the prompt text E_p and the video features E_v are combined and fed into the LLM to obtain the predicted caption v.

$$T = Decoder(Concat(E_p, E_v)). \tag{4}$$

3.2. Semantic feature extractor

Densely sampled frames often contain a significant amount of redundancy and potentially irrelevant information, which can negatively impact overall performance. Thus, efficiently and accurately representing videos within our proposed framework presents a substantial challenge. To overcome this hurdle, we introduce a Semantic Feature Extractor that incorporates an attention mechanism in tandem with

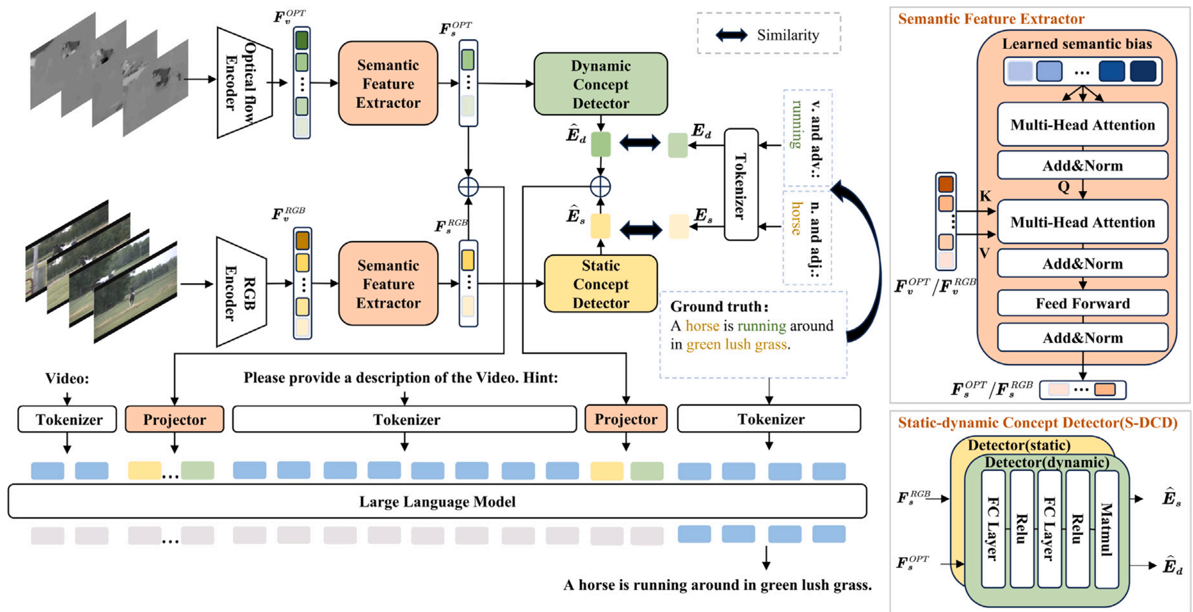

Fig. 1. Overall of the proposed DiSCo. DiSCo incorporates two branches for extracting static and dynamic semantic features separately via two semantic feature extractors. Utilizing these extracted semantic features, DiSCo introduces a static-dynamic concept detector to predict concept vectors. Finally, in conjunction with prompts, the resulting information is fed into the large language model. Encoder and decoder parameters are frozen during training.

learnable vectors referred to as semantic biases. This approach aims to sift through the video's content, focusing on pertinent details while filtering out unnecessary noise.

The structure depicted in Fig. 1 corresponds to a transformer decoding layer. In addition, we introduce a semantic bias $q_{bias} \in \mathbb{R}^{N_{bias} \times d_E}$ referred to as the Query, while the video features F_v serve as the Key and Value inputs for the transformer decoding layer, facilitating cross-attention computation. The corresponding formula is presented below.

$$F_s = Extractor\left(q_{bias}, F_v\right) = TD\left(q_{bias}, F_v\right), \tag{5}$$

where TD represents the decoder layer of transformer [19]. For detailed configuration, please refer to Fig. A.1. Notably, due to the two-stream structure, RGB features F_v^{RGB} and optical flow features F_v^{OPT} should be separately passed through the extractor to obtain the semantic features, and then they are fused to obtain the final semantic features $F_s \in \mathbb{R}^{(N_{bias1}+N_{bias2}) \times d_E}$.

3.3. Static-dynamic concept detector

The key concepts in the video description should be used as hints for caption generation. Building upon the baseline, we propose a branch that utilizes RGB features and optical flow features to detect static and dynamic concepts respectively. Specifically, a set of N_c words associated with static concepts (such as nouns and adjectives) and another set of words related to dynamic concepts (such as verbs and adverbs) are extracted from ground truth and then embedded into vectors representing static concepts $E_s \in \mathbb{R}^{N_c \times d_D}$ and dynamic concepts $E_d \in \mathbb{R}^{N_c \times d_D}$ respectively. It is worth noting that if there are fewer relevant words in the GT captions than N_c, zero vectors are added to complete them; whereas if there are more than N_c relevant words, only the top N_c words are considered. Leveraging the previous two-stream encoder architecture, the static concept detector corresponds to the RGB encoder while the dynamic concept detector corresponds to the optical flow encoder. These detectors predict related static concept vectors $\hat{E}_s \in \mathbb{R}^{N_c \times d_D}$ and dynamic concept vectors $\hat{E}_d \in \mathbb{R}^{N_c \times d_D}$ based on semantic features. The formula can be expressed as follows.

$$\hat{E}_d, \hat{E}_s = D - SDetector\left(F_s^{RGB}, F_s^{OPT}\right), \tag{6}$$

where $D - SDetector$ consists of two independent MLPs that take inputs F_s^{OPT} and F_s^{RGB} to predict static and dynamic concept vectors, respectively. During training, we aim to minimize the cosine similarity function $L_{concept}$.

3.4. Training details

The Semantic Feature Extractor and Static-Dynamic Concept Detector, proposed above, are incorporated into the baseline model. The Vicuna-7B [20] model is employed as the decoder of the baseline. The dimension of the semantic bias q_{bias} is set to be w times the dimension of F_v, where w represents the compression rate. Our training objectives encompass caption generation L_{cap} and concept detection $L_{concept}$. During training, we freeze the parameters of the encoder and decoder while optimizing the rest of the model using a comprehensive loss function $L = L_{cap} + L_{concept}$.

4. Experiments

In this section, we first introduce the experimental setup. After that, in order to show the effectiveness of our proposed framework, we trained our model on two commonly used datasets and obtained top-ranked results. Finally, we have the ablation experiment.

4.1. Experimental setup

Dataset. We conduct experiments on two benchmark datasets: MSVD [21] and MSR-VTT [22]. MSVD contains 1970 videos of about 10 s, each with up to 45 captions. MSR-VTT is a diverse open domain dataset containing 10,000 videos with 20 captioning per video. We have 6513, 497 and 2990 videos for training, validation and testing, respectively, according to the official division.

Metrics. MS COCO Caption Evaluation [23] is used to compute four commonly used metrics on two datasets: CIDEr [24], BLEU@4[25], METEOR [26], and ROUGE-L [27].

Features. Kinetics [28] pretrained Video Swin Transformer [6] is used to extract RGB features, and mmaction2-based BN-InceptionNet from TSN [7] pre-trained on Kinetics to extract optical flow features.

Table 1
Comparison with previous work on test sets of MSRVTT and MSVD. For those competing methods, we extract their performance from their latest version of paper. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

Model	Year	Auxiliary Methonds	MSRVTT				MSVD			
			C	B@4	M	R	C	B@4	M	R
SGN [31]	2021	–	49.5	40.8	28.1	–	94.3	52.8	35.5	72.9
MGCMP [32]	2021	–	51.4	41.7	28.9	62.1	98.5	55.8	36.9	74.5
HRNAT [9]	2022	–	48.2	42.1	28.0	61.6	98.1	55.7	36.8	74.1
TVRD [33]	2022	–	51.8	43.0	28.7	62.2	84.3	50.5	34.5	71.7
RSFD [4]	2023	–	53.1	43.4	29.3	62.3	96.7	51.2	35.1	72.9
ORG-TRL [34]	2020	Object	50.9	43.6	28.8	62.1	95.2	54.3	36.4	73.9
LSRT [5]	2022	Object	49.5	42.6	28.3	61.0	98.5	55.6	37.1	73.5
OpenBook [12]	2021	Retrieval	52.3	43.1	29.0	61.9	–	–	–	–
R-ConvED [10]	2022	Retrieval	48.3	37.8	27.6	–	83.4	50.5	34.4	–
STA-FG [2]	2020	Concept	–	20.8	27.4	–	–	52.7	34.5	–
MAD-SAP [3]	2020	Concept	48.5	41.3	28.3	61.4	90.8	53.3	35.4	72.0
TTA [30]	2021	Concept	46.7	41.4	27.7	61.1	87.7	51.8	35.5	72.4
Ours	2024	Concept	**55.7**	**44.0**	**29.4**	**62.7**	**111.8**	**57.4**	**38.5**	**74.8**

Table 2
A comprehensive study of our model's components compared to the baseline method described in Chapter 3.1 on MSRVTT dataset. SFE stands for Semantic Feature Extractor, S-DCD for Static-Dynamic Concept Detector, and Optical Stream refers to the optical flow branch. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

Model	SFE	S-DCD	Optical stream	C	B@4	M	R
Baseline	–	–	✓	51.6	42.3	28.2	61.3
Ours	–	✓	✓	54.1	43.5	28.9	62.0
	✓	–	✓	54.9	43.2	29.6	62.3
	✓	✓	–	54.1	41.5	29.1	61.1
	✓	✓	✓	**55.7**	**44.0**	**29.4**	**62.7**

Table 3
Performance comparison of the baseline and proposed model across different numbers of input video frames (T) on MSRVTT dataset. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

T	Model	C	B@4	M	R
16	Baseline	48.1	40.2	27.7	60.1
16	Ours	**52.7**	**41.0**	**28.0**	**60.9**
32	Baseline	49.8	41.9	27.9	60.3
32	Ours	**54.5**	**42.2**	**29.1**	**61.8**
48	Baseline	50.4	41.9	28.1	61.0
48	Ours	**54.8**	**43.1**	**28.9**	**62.1**
64	Baseline	51.6	42.3	28.2	61.3
64	Ours	**55.7**	**44.0**	**29.4**	**62.7**

Detailed architectures and parameter configurations of these models are provided in Appendix A.

Training details. We trained the model on an NVIDIA TITAN RTX with batch size 4, learning rate 3e-5 and AdamW optimizer [29] for 3 epochs. Each training epoch required approximately 8 h to complete.

4.2. Quantitative results

Table 1 presents a meticulous comparison of results on both the MSVD and MSRVTT datasets. It is evident that our proposed method surpasses previous approaches, delivering higher evaluation metrics across the board, with a particularly pronounced improvement in the CIDEr score—a metric specifically engineered to gauge the descriptive quality and relevance of visual content. Notably, while not requiring the construction of any additional vocabulary as compared to concept-detection-based methods [2,3,30], our approach nonetheless outperforms them, thereby substantiating the superior effectiveness and robustness of our integrated concept detection technique.

4.3. Ablation study

The contribution of each component was further investigated through additional ablation studies conducted on the MSRVTT dataset, as summarized in Table 2.

The effect of video frames. To assess our model's robustness across diverse video frame scenarios, we systematically trained and evaluated it using uniform frame sets (T = 16, 32, 48, 64) per clip, benchmarking against a baseline. The results on the MSRVTT dataset, as shown in Table 3, highlight that integrating SFE and S-DCD consistently boosts performance, regardless of frame configuration. Notably, an increase in frame counts leads to significant metric enhancements, underscoring the critical importance of denser visual information in elevating video captioning capabilities.

Table 4
Impact of the compression rate (w) within the SFE module on model performance on MSRVTT dataset, with w ranging from 0 to 1. Comparison includes a model variant without the SFE module (denoted by "w/o SFE"). Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

Model	w	C	B@4	M	R
Ours	1	54.9	43.7	28.8	62.0
	0.75	55.4	44.0	29.3	62.3
	0.5	55.0	**44.2**	**29.4**	62.6
	0.25	**55.7**	44.0	**29.4**	**62.7**
w/o SFE	–	54.1	43.5	28.9	62.0

The effect of SFE. Table 2 outlines ablation studies that confirm the SFE's impact. The baseline model, devoid of both SFE and S-DCD, contrasts sharply with the SFE-only model, clearly demonstrating the SFE's singular contribution to performance enhancement. Delving further into assessing SFE's efficacy, Table 4 explores the influence of varied compression rates (w). Models that integrate SFE, operating across a range of w from 0.25 to 1, consistently outperform those lacking SFE, thus emphasizing the SFE's pivotal role within our architecture. Striking an optimal balance between efficiency and efficacy, we settle on a compression rate of w = 0.25. This finding underscores that an abundance of video features does not necessarily equate to superior model performance; indeed, extraneous information within the video can potentially degrade model performance.

The effect of S-DCD.

Table 2 highlights a marked performance leap in models utilizing only S-DCD, versus the baseline, and demonstrates S-DCD's effectiveness through performance dips upon removal. Table 5 reveals a direct correlation between enhanced performance and the utilization of concept vectors, with high similarity scores between predicted vectors and ground truths correlating with superior metric values. Poor performance is typically associated with low vector-content similarity. These

Table 5
Impact of predicted semantic vector accuracy on model performance, categorized into 10%-best (top 10% similarity to ground truth) and 10%-worst (bottom 10% similarity to ground truth), was evaluated on the MSRVTT dataset. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

Similarity	C	B@4	M	R
10%-best	184.0	71.1	48.0	84.7
10%-worst	51.8	37.6	29.1	62.4

Table 6
Impact of varying the number of predicted Concepts by S-DCD on MSRVTT. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

N_{words}	C	B@4	M	R
2	54.4	42.0	29.1	61.6
4	**55.7**	**44.0**	**29.4**	**62.7**
6	54.9	43.2	28.7	61.8

Table 7
Effect of varying the structure of the S-DCD module on model performance on the MSRVTT dataset. d_1 and d_2 are both dimensions of the encoder's hidden layers. Performance metrics used include CIDEr (C), BLEU@4 (B@4), METEOR (M), and ROUGE (R).

Architecture	C	B@4	M	R
(d_1, d_2)	55.3	43.9	29.3	62.5
$(d_1, d_2)(d_2, d_2)$	54.3	**44.0**	29.2	62.0
$(d_1, d_2*2)(d_2*2, d_2)$	**55.7**	**44.0**	**29.4**	**62.7**
$(d_1, d_2*4)(d_2*4, d_2)$	54.3	42.5	28.7	61.7

findings underscore S-DCD's critical role in the model's architecture. Further investigations into the number of concepts and model structure, as shown in Tables 6 and 7, indicate optimal results with N_{words} = 4 concept vectors. Fewer predicted concepts can lead to incomplete information, whereas an excess of predicted concepts may introduce noise due to lower content correlation. The structure $(d_1, d_2 * 2)(d_2 * 2, d_2)$ excels, showcasing the importance of balanced vector usage and structural design.

The effect of optical flow. To verify the effectiveness of the optical flow branch, ablation studies in Table 2 reveal a significant performance drop across video captioning metrics upon its removal.

4.4. Qualitative results

To provide a more intuitive understanding of the advantages of concept-driven video captioning, we utilized heatmaps to visually depict the closest word associations with detected concept vectors, as illustrated in Fig. 2. DiSCo effectively discerns pertinent concepts linked to the video content through its integration of S-DCD. For instance, in example (a), DiSCo accurately identifies key elements such as "woman", "man", "dog", and "walking". Likewise, in scenario (b), it detects relevant concepts like "man", "glass", and "talking", while in case (c), it successfully recognizes "cartoon" and "dancing". The identified concepts act as informative prompts that significantly aid in reducing hallucination occurrences, thereby contributing to the generation of scholarly precise and comprehensive captions. In comparison, both the baseline model and the one without S-DCD demonstrate misinterpretations; for instance, they fail to correctly describe (a) as "a couple", whereas DiSCo accurately captures this detail. In (c), the non-S-DCD model notably suffers from a substantial hallucination issue, mistakenly associating 'video game' content, whereas DiSCO consistently produces more accurate descriptions. Moreover, examples (b) and (d) underscore how DiSCo furnishes more detailed and holistic accounts of the videos. It also demonstrates prowess in recognizing embedded textual information within videos, as seen in (b), where it identifies the term "war", which is instrumental in generating an accurate video caption. This capability further underscores the effectiveness and versatility of the DiSCo model in enhancing the accuracy and fidelity of video captioning tasks.

5. Conclusions

We introduce DiSCo, a concept-driven video captioning model underpinned by the power of a LLM. The framework incorporates a Semantic Feature Extractor (SFE) and a Static-Dynamic Concept Detector (S-DCD). These components are responsible for filtering out semantically irrelevant features extracted by the visual model and identifying key concepts crucial for generating captions. This design effectively addresses the issue of redundant features present in mainstream methods, thereby enhancing the accuracy of caption generation. Additionally, by freezing the parameters of the pre-trained models and training only the SFE and S-DCD, an efficient and targeted optimization strategy is achieved. During training, concept prediction serves as an auxiliary task, helping the model to focus more on semantically relevant features. At inference, these concepts guide the language model, providing robust cues to reduce common hallucinations in generative models. Empirical evaluations on the MSVD and MSR-VTT datasets demonstrate that DiSCo significantly outperforms existing methods in improving the quality and relevance of generated captions.

Looking ahead, several directions warrant further exploration and development. First, improving algorithms to enhance the accuracy of concept prediction is essential for providing the model with more precise prior knowledge. Second, given the high computational demands of optical flow processing, developing more efficient data processing techniques will be crucial for optimizing the model's runtime efficiency. Third, integrating additional sensory inputs, such as audio or depth information, can enrich feature representation to provide a more nuanced description of video content.

CRediT authorship contribution statement

Xin Ren: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yufeng Han:** Writing – review & editing, Methodology, Conceptualization. **Bing Wei:** Writing – review & editing, Validation, Formal analysis. **Xue-song Tang:** Methodology, Formal analysis, Conceptualization. **Kuangrong Hao:** Supervision, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kuangrong Hao reports financial support was provided Shanghai Pujiang Program. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Architecture of subnetworks

In this section, we show architectures of the three subnetworks in our DiSCo framework and present additional details of the network structures and configurations.

RGB Encoder. The RGB encoder backbone to our DiSCo model is the Video Swin Transformer [6], more precisely the Base version pretrained on Kinetics 400 [28]. While the original pipeline is designed for action classification, in our context of video description, we have

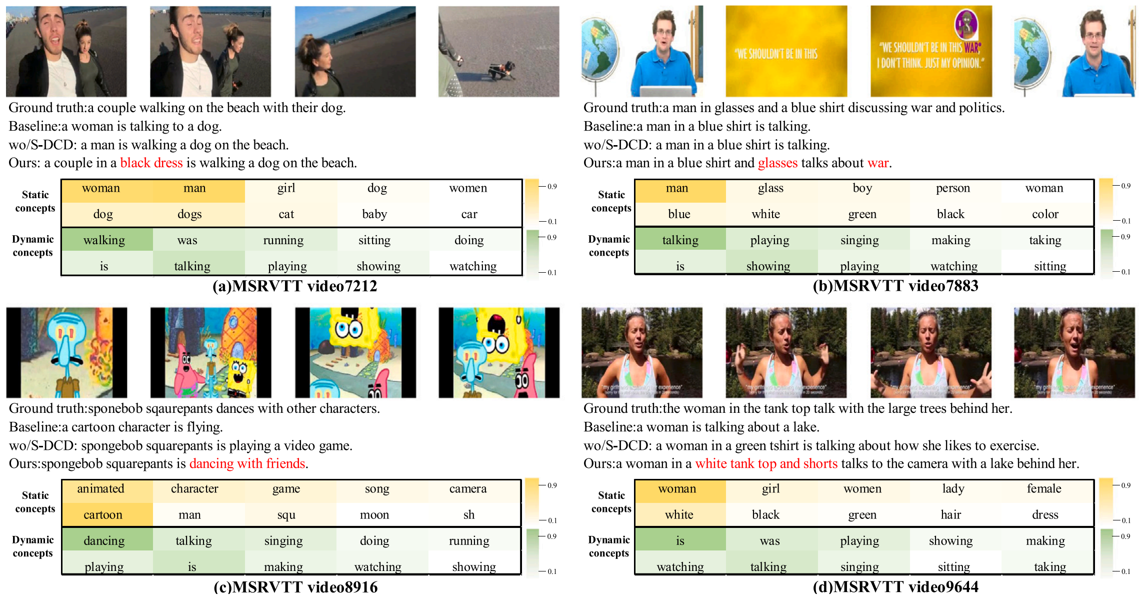

Fig. 2. Qualitative examples generated by DiSCo. Each example consists of a manually annotated ground-truth caption, captions generated by Baseline, the proposed DiSCo model, DiSCo model without S-DCD, and a heatmap highlighting the words most similar to the detected concept vector. The detected concepts are categorized into static and dynamic concepts. Precise keywords in the subtitles are emphasized in red.

Table A.1
The configuration and parameters of each module.

Video Swin Transformer [6]		TSN [7]		SFE		Vicuna-7B [20]	
Parameter	Value	Parameter	Value	Parameter	Value	Parameter	Value
Patch size	(2, 4, 4)	Input modality	Optical flow	Hidden size	1024	Hidden size	4096
Depths	[2, 2, 18, 2]	New length	2	Num heads	8	Top_p	0.95
Num heads	[4, 8, 16, 32]	Dropout ratio	0.7	FFW hidden size	2048	Temperature	1
Window size	(8, 7, 7)			Dropout ratio	0.1	Num_beams	5
				Num queries	8		

modified the network as illustrated in Fig. A.1(a). we decode the video into frames and sparsely sample 64 frames. These sampled video frames are then fed into the RGB encoder to extract RGB features $F_v^{RGB} \in \mathbb{R}^{32\times1024}$. More details on the specific configuration and parameters of the Video Swin Transformer [6] are listed in Table A.1.

Optical Flow Encoder. The optical flow encoder backbone to our DiSCo model is the mmaction2-based BN-InceptionNet from TSN [7] which is pretrained on Kinetics 400 [28]. Fig. A.1(b) shows the structure of the optical flow encoder and Table A.1 presents the configuration and parameters. We uniformly distribute the flow maps into 32 clips per video, sample 2 frames per clip, and obtain a 1024-dimensional feature vector for each clip, resulting in optical flow feature $F_v^{OPT} \in \mathbb{R}^{32\times1024}$.

Decoder. The decoder in DiSCo is vicuna-7B [20]. Fig. A.1(c) illustrates the architecture of the vicuna-7B [20], while Table A.1 details its configuration and parameters. We concatenate the prompt "video:<VideoHere>. Please provide a detailed description of the video. Hint:<ConceptHere>" with the semantic features and concept features, and input them into the decoder. The placeholders <VideoHere> and <ConceptHere> will be replaced with semantic features and concept features, respectively. It is important to note that both the semantic features and concept features need to be passed through a mapping layer to ensure their dimensions match those of vicuna-7b's hidden layers.

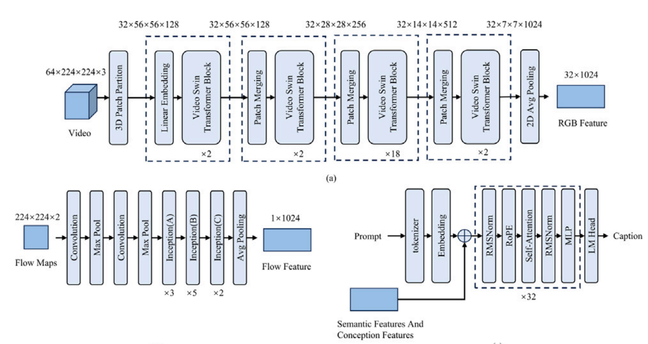

Fig. A.1. Architecture of subnetworks. (a)Detailed architecture of our RGB encoder model based on the Video Swin Transformer [6]. (b) Detailed architecture of our optical flow encoder model based on the TSN [7]. (c) Detailed architecture of our decoder model based on the vicuna-7B [20].

Data availability

Data will be made available on request.

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